MIL-C-11732D
20 March 1973
SUPERSEDING
MIL-C-11732C
30 April 1964

MILITARY SPECIFICATION

CYLINDER, COMPRESSED GAS: CHLORINE;

DOT 3A480 OR DOT 3AA480,

GENERAL SPECIFICATION FOR

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers new steel cylinders, with valve and cap, for chlorine service.

1.2 Classification. The classification for the cylinder shall include the base specification letter and number, the slash number for the applicable specification sheet and the dash number indicating size for the cylinder as specified (see 6.2).

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

BB-N-411	- Nitrogen, Technical.
PPP-B-601	- Boxes, Wood, Cleated-Plywood.

Military

MIL-V-2	- Valve, Cylinder, Gas (for Compressed or Liquefied Gases), General Specification for.
MIL-V-2/20	- Valve, Cylinder, Gas: Chlorine Outlet 661.

FSC 8120

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MIL-T-704	- Treatment and Painting of Materiel.
MIL-C-17376	- Cap, Compressed-Gas Cylinder.

STANDARDS

Military

MIL-STD-101	- Color Code for Pipelines and for Compressed Gas Cylinders.
MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-147	- Palletized and Containerized Unit Loads, 40" x 48" Pallets, Skids, Runners, or Pallet-Type Base.
MS39218	- Flange Attachment, Compressed Gas Cylinder: Neck Ring.

DRAWING

SH

810-1385867	- Decalcomania for Navy Gas Cylinders.

(Copies of specifications, standards, and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC., AGENT

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Associations, Inc., ATTN: Tariff Order Section, 1616 P Street, NW, Washington, DC 20036.)

UNIFORM CLASSIFICATION COMMITTEE, AGENT

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, ATTN: Tariff Publishing Officer, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

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DEPARTMENT OF TRANSPORTATION (DOT)

Code of Federal Regulations - Title 49, Transportation

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

NATIONAL BUREAU OF STANDARDS

Handbook H28 - Screw-Thread Standards for Federal Services.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

3. REQUIREMENTS

3.1 Description. Cylinders shall conform to DOT Hazardous Material Regulations Code of Federal Regulations, Title 49, Specification 3A or 3AA, shall be as specified in the applicable specification sheet and as specified herein. Service pressure shall be 480 psi.

3.2 Construction. The cylinders shall be true of form and shall be stable when in an upright position. The neck of each cylinder shall be tapped internally 3/4-14NGT (Cl)-1. Threads shall conform to Handbook H28.

3.3 Cylinder processing.

3.3.1 Internal surfaces. The internal surfaces of drawn cylinders, prior to necking, shall be free of mill scale, burrs, oxidation products and all foreign matter. When required, only iron-base abrasives shall be used to remove unstable surfaces or oxidation products from the cylinder wall. Metal discoloration, rust bloom, and fixed scale shall not be considered a criteria for cylinder rejection.

3.3.2 Cylinder cleaning. After necking, drilling, threading and deburring, the cylinders shall be cleared of loose particles by vacuum or by inverting the cylinder and use of a compressed air sweep (see 6.3). The cylinder shall be flushed with hot fluids when necessary to remove grease, oil, and fat from processing. Nonvolatile hydrocarbons shall not exceed a residual of one milligram per square foot of internal cylinder area. Chemicals that will react with chlorine shall not be used.

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3.3.3 Cylinder drying. Internal cylinder moisture from processing or hydrostatic testing shall be removed by inverting and steaming the surfaces until hot to the touch, followed by drying with filtered air having a dew-point less than 30° F. In lieu of filtered air, drying by suction of ambient air through the cylinder is allowed. The residual air in the dry cylinder shall be replaced by introducing dry nitrogen gas conforming to BB-N-411, grade B, through a bayonet extending to the bottom end of the cylinder. Closure shall be carried out immediately to limit atmospheric interchange.

3.4 Closure. Cylinders shall be equipped with a flange conforming to MS39218 and a cap conforming to MIL-C-17376 as specified in the applicable specification sheet. A plug or a closed valve conforming to MIL-V-2/20, as specified (see 6.2) shall be installed in the neck of the cylinder.

3.5 Tare weight. The tare weight of the cylinder shall be the weight of the cylinder with valve but excluding the valve protection cap. The weight shall be expressed in pounds and quarter pounds expressed in 4 ounce increments to the nearest quarter pound. The figures expressing pounds shall be separated from the figures expressing ounces by a dash (-).

3.6 Test date. The cylinder shall be delivered within 1 year of the test date stamped on the cylinder.

3.7 Cylinder identification. Unless otherwise specified herein, marking shall be not less than 1/4 inch high.

3.7.1 Standard markings. Standard markings shall be plainly and permanently marked by stamping on the shoulder of each cylinder starting near the neck, as follows:

(a) "DOT 3A480" or "DOT 3AA480" in characters not less than 3/8 inch high for cylinders more than 6 inches in outside diameter.
(b) Serial number (see 6.4), as registered with the Bureau of Explosives in characters not less than 1/2 inch high for cylinders more than 6 inches in outside diameter.
(c) "USGOVT", Government symbol as registered with the Bureau of Explosives in characters not less than 1/2 inch high for cylinders more than 6 inches in outside diameter.
(d) The inspector's official mark.
(e) Date of testing diametrically opposite the above marking.
(f) Manufacturer's or retester's mark near date of tests, preferably directly above it.
(g) The tare weight shall be stamped in a position other than in sequence with the serial number or test date.

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3.7.2 Additional markings. In addition to the above markings, the cylinders shall bear the following markings:

(a) Name of gas stenciled on the cylinder in accordance with MIL-STD-101.
(b) When specified (see 6.2), two decalcomanias conforming to Drawing 810-1385867 shall be affixed diametrically opposite one another on the sides of the cylinder, 90 degrees from the stenciled name of the gas.

3.8 Treatment and painting. Each cylinder and cap shall be treated and painted externally in accordance with MIL-T-704, type A, color conforming to the cylinder color code requirements of MIL-STD-101.

3.9 Workmanship. Cylinders, valves, plugs, flanges, and caps shall be clean and free from grit, fins, pits, and loose scale. Edges shall be rounded and chamfered. Cylinders shall be cleaned and free of dents, scratches, and any other surface defects detrimental to the intended use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. The supplier is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards.

4.2 Classification of inspection. Inspection shall be classified as follows:

(a) Quality conformance inspection (see 4.3).
(b) Inspection of preparation for delivery (see 4.5).

4.3 Quality conformance inspection.

4.3.1 Examination. Each cylinder shall be examined in accordance with 4.4.1. The presence of any defect shall constitute failure for the cylinder.

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4.3.2 Tests. All tests shall be completed within 1 year from date of acceptance by the Government.

4.3.2.1 Lot. A lot shall consist of not more than 200 cylinders produced from the same heat of steel, having the same treatment cycle and offered for delivery at the same time.

4.3.2.2 Sampling. Sampling for tests shall be in accordance with MIL-STD-105, inspection level S II.

4.3.2.3 Samples. Samples selected in accordance with 4.3.2.2 shall be tested as specified in 4.4.2. AQL shall be 2.5 percent defective.

4.4 Inspection procedure.

4.4.1 Examination. Cylinders shall be examined as specified herein for the following defects:

101. Cylinders not in accordance with DOT specification 3A or 3AA.
102. Construction not as specified.
103. Closure not as specified.
104. Dimensions not as specified.
105. Tare weight not as specified.
106. Cleaning and drying not as specified.
107. Cylinders not purged with nitrogen as specified.
108. Marking not as specified.
109. Treatment and painting not as specified.
110. Test date more than 1 year prior to date of acceptance by the Government.

4.4.2 Test.

4.4.2.1 Nonvolatile hydrocarbons. Place a clean cork in the cylinder neck and clean the area around the cork and cylinder neck thoroughly with redistilled chlorinated hydrocarbon solvent and wipe dry with a clean rag. Remove the cork and pour 300 milliliters (ml) of redistilled chlorinated hydrocarbon solvent into cylinders with up to 3 square feet of internal area. For larger cylinders, add an additional 100 ml for each square foot of internal area over 3 square feet. Replace the cork. Lay the cylinder on its side, and roll the cylinder through 360 degrees on a level surface for 10 minutes. Remove the cork from the cylinder and pour the solvent-extract into a clean 500 ml graduated cylinder. The solvent-extract volume shall be recorded and analyzed for nonvolatile hydrocarbon as follows:

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a. Evaporation Method.

(1) Weigh two 600 ml beakers to the nearest 0.1 milligram, record and identify as A and B.
(2) Place the solvent-extract in beaker A, and place an amount of base solvent equal to the volume reclaimed from the test cylinder in beaker B as a control standard.
(3) Place both beakers A and B in a hooded boiling water bath or in a ventilated oven at 100°C and evaporate until dry but for not less than 1 hour. Cool the beakers to room temperature, weigh to the nearest 0.1 milligram and record.
(4) Subtract the beaker tare weight from the weight of the beakers and residues respectively, to determine the weight of residue A from the solvent-extract and the weight of residue B to determine the weight of any residue inherent in the base solvent.
(5) Substract the weight of residue B from the weight of residue A to determine the total weight of nonvolatile hydrocarbon residue and divide by the number of square feet of surface area. Nonvolatile hydrocarbons in excess of 1 milligram per square foot shall constitute failure of this test.

b. Infrared spectrophotometer. A sample of solvent-extract shall be analyzed against a reference standard of base solvent with a known hydrocarbon level of 1 milligram per 100 ml. A response in a functional range displaying greater contamination of nonvolatile hydrocarbons in the solvent-extract than found in the reference standard of 1 milligram per 100 ml shall constitute failure of this test.

c. In case of dispute final determination shall be made by the evaporation method.

4.5 Inspection of preparation for delivery.

4.5.1 Sampling. Sampling shall be in accordance with MIL-STD-105.

4.5.2 Examination. Samples selected in accordance with 4.5.1 shall be examined for the following defects. AQL shall be 4.0 percent defective.

111. Petroleum base solvents, preservatives, or lubricants applied to cylinders.
112. Boxes exceeding 250 pounds gross not modified with skids.
113. Strapping not zinc coated for level A.
114. Palletization not as specified.
115. Markings illegible, incomplete, or incorrect.

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5. PREPARATION FOR DELIVERY

5.1 Preservation. No preservatives, of any kind, shall be applied to any part of the cylinders.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

5.2.1 Level A. Cylinders segregated for size and type shall be packed in strapped pallet loads in accordance with load type IX, or storage aid 25 of MIL-STD-147 except that the cap end of the cylinder shall not extend beyond the pallet or space edge. Cylinders smaller than 30 inches in length (not including valve) shall be palletized as specified herein or shall be packed in close-fitting boxes conforming to PPP-B-601, overseas type, style 1, up to the weight limitation of the container. Strapping shall be zinc coated.

5.2.2 Level B. The cylinders shall be palletized or boxed as specified for level A except boxes shall be domestic type and the strapping need not be zinc coated.

5.2.3 Level C. The cylinders shall be prepared for shipment in a manner to assure damage free delivery from supplier to initial destination at the lowest ratings in compliance with Uniform Freight Classification rules or National Motor Freight Classification rules.

5.3 Marking (military agencies). Loose cylinders and pallets shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. Cylinders are to be used for the shipment and storage of chlorine.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Title, number, and date of applicable specification sheet.
(c) Cylinder classification (see 1.2).
(d) Whether a plug or valve is required (see 3.4).
(e) When decalcomanies are required (see 3.7.2(b)).
(f) Level of packing required (see 5.2).

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6.3 Loose particulate matter in stored cylinders. Cylinders that have been cleaned, dried, valved or plugged and subjected to storage and handling within delivery time limits will be acceptable for chlorine service if accumulated loose ferro-related particulate matter resulting from handling does not exceed 1 gram per 1500 cubic inches of cylinder capacity.

6.4 Serial numbers. Cylinder serial numbers shall be prefixed by a two or three letter symbol designating the procuring agency and shall have a two letter suffix designating the supplier furnished by the contracting officer. The serial numbers shall be assigned by the supplier. These numbers may be consecutive with the supplier's regular production numbers or of a series established specifically for customer cylinders. However, all cylinders on a given contract will be numbered consecutively and controls will be exercised to preclude duplication on future deliveries to the Government.

Custodians:

Army - ME
Navy - SH
Air Force - 68

Review activity:

Navy - MC

User activities:

Army - MJ
Navy - YD

Preparing activity:

Army - ME

Project No. 8120-0268

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

Form Approved
Budget Bureau No 22-R255

INSTRUCTIONS This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost Comments and the return of this form will be appreciated Fold on lines on reverse side staple in corner and send to preparing activity Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements

SPECIFICATION MIL-C-11732D, Cylinder, Compressed Gas: Chlorine; DOT 3A480 or DOT 3AA480, General Specification For

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1 HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A GIVE PARAGRAPH NUMBER AND WORDING

B RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2 COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3 IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes" in what way?)*

4 REMARKS *(Attach any pertinent data which may be of use in improving this specification If there are additional papers attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity Optional)*

DATE

DD FORM 1426
11 Oct 72 (Rev)

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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